

Channel selection improves accuracy for pediatric users of motor imagery brain-computer interfaces

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Abstract—Children are an under-served population in the field of brain-computer interface (BCI) development. The high prevalence of lifelong disability coupled with the diversity and plasticity of children's brains make them ideal candidates for personalized BCI systems. Channel selection methods provide a tool for the in-session personalization of BCI systems. To evaluate the efficacy of channel selection for pediatric users, we tested four wrapper-based channel selection algorithms, sequential forward selection (SFS), sequential backward selection (SBS), sequential forward floating selection (SFFS), and sequential backward floating selection (SBFS) on offline motor imagery BCI data from three datasets involving typically developing children. The purpose was to assess the performance benefits and computational costs of each algorithm. All algorithms provided classification accuracy gains of 10-15 % with their optimal subsets. The time required to reach the optimal subsets varied between algorithms, but all took less than 80 s with mean completion times of 9.5 s and 35.8 s for the fastest (SFS) and slowest (SFFS), respectively. Adjusting the stopping criterion of the algorithm enables users to further reduce computation time with a disproportionately small effect on classification accuracy. All methods demonstrated an ability to prioritize expected physiological regions of interest and leave out channels detrimental to the classifier. Channel selection offers personalization of the BCI system for a specific user and a specific classifier. These findings emphasize the value of using personalized channel selection algorithms to improve motor imagery BCI systems for pediatric users.

Index Terms—brain-computer interface, BCI, motor imagery, channel selection, wrapper methods, SFS, SBS, SFFS, SBFS

I. INTRODUCTION

Brain-computer interfaces (BCI) are systems which translate brain activity into inputs for a computer [1]. Brain activity is commonly measured using electroencephalography (EEG) due to its high temporal resolution and relatively low cost [2]. EEG headsets vary widely in the number of channels that they support. Minimal consumer-grade headsets may have only 8 channels, like the Neurosity Crown (Neurosity, Brooklyn, USA). Research-grade systems may have significantly more, like the 1024 channel g.tec gPANGOLIN (g.tec medical engineering GmbH, Austria). The diversity of these systems reflects many factors, including the diversity of their users. BCIs can be an effective tool for improving quality

of life by restoring or augmenting the abilities of individuals with amyotrophic lateral sclerosis (ALS) or spinal cord injury (SCI) [1]. Additionally, BCIs have great potential to improve the quality of life of children with severe neurological disabilities, such as those with quadriplegic cerebral palsy (QCP), by helping them to realize their rights to communication, independence, and life participation.

Building BCIs for children may require a more individualized approach as their brains exhibit physiological, connective, and developmental differences compared to adults, particularly in the case of early life injury [3]. Children, including those with QCP, can use BCI systems [4]–[8]. However, these systems often require a high degree of personalization to be functional for users with QCP [7]. Achieving good signal quality across all channels can be a challenge for individuals with different head shapes and brain structure [9]. With these differences, children, especially those with early life injury or other complex needs, require specific attention in BCI design and implementation. This need has not been adequately addressed. A systematic review of pediatric BCI studies in 2021 found only 12 studies which involved children, most of which only involved typically developing children [10]. More work must be done to improve BCI systems for pediatric users.

Motor imagery (MI) BCI is a paradigm which exploits changes in sensorimotor rhythms (SMR) as the user-modulated signal to control a BCI system [11]. This is done through the imagined motion of one or more parts of the body [11]. A classic example is to imagine opening and closing one's left or right hand. When the action is imagined, it causes an identifiable event-related desynchronization (ERD) over the corresponding area of the motor and pre-motor cortices, e.g. the contralateral hand area of the motor cortex [12]. Because these signals are localized to the motor regions of the brain, some centrally located channels, such as C3 and C4, are more likely to capture relevant signal changes. However, like all EEG-based BCI systems, MI signals are susceptible to environmental and physiological noise due to their low signal-to-noise ratio (SNR) [11]. Locally distributed signal and more uniformly distributed noise results in varying SNR between channels. Channels with the greatest SNR are assumed to be most beneficial to the classifier.

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TABLE I
DATASET CHARACTERISTICS

Dataset	# Participants	Age Range (Mean)	# Female	# Channels	Channel Labels	Epoch Length (s)	# Epochs
DS1	32	7-16 (10)	20	19	C3, C4, Cz, F3, F4, F7, F8, Fp1, Fp2, Fz, O1, O2, P3, P4, Pz, T3, T4, T5, T6	2	108
DS2 & DS3	27	5-17 (11)	11	16	C1, C2, C3, C4, C5, C6, CP1, CP2, CP3, CP4, CPz, Cz, FC3, FC4, FCz, Pz	1.5	120

Channel selection is a method used to identify subsets of channels which, when classified, perform better than the set of all channels. In this way, channel selection seeks to remove noisy channels while maintaining channels with relevant information. Since each channel contains information which is used for classification, channels can themselves be considered as features and benefit from the application of feature selection methods. To be precise, we will refer to these feature selection methods applied to EEG channels as channel selection methods. Wrapper methods are a subset of channel selection methods which follow an iterative process to try different combinations of channels and test them using a target classifier [13]. The alternatives to wrapper methods are filter methods, which use available EEG data to estimate an optimal channel set. Wrapper methods generally provide subsets which are closer to optimal, because they select channels relevant to an individual user and an individual classifier, but they are often criticized for being slow and computationally expensive [13], [14]. However, this is only detrimental if the classifier itself is computationally expensive. Wrapper methods have been employed with Riemannian geometry classifiers because of their low computational cost, and state-of-the-art-performance when limited data is available [15], [16]. There have also been attempts to make wrapper methods faster by grouping channels together and adding/removing them based on their combined effect [17].

In this work, we evaluated four sequential search wrapper algorithms for their ability to offer personalization and improved performance for pediatric users of BCIs. The tested algorithms included sequential forward selection (SFS) [18], sequential backward selection (SBS) [19], sequential forward floating selection (SFFS) [20], and sequential backward feature selection (SBFS) [20]. All four of these algorithms are recognized as effective, but do not guarantee optimality [21]. SFS and SBS are relatively fast and simple to implement, but suffer from the nesting effect, where once a feature has been added or removed the change is permanent [21]. SFFS and SBFS remove the nesting effect at the cost of greater computational time [21]. SFS [22]–[24], SBS [24], SFFS [17], [25]–[27], and SBFS [28] all have precedent in being applied to BCI channel selection, but to the authors' knowledge they have never been compared and none have been studied in children. Based on our review of the literature, the forward sequential methods are more common than their backward counterparts. For each algorithm we evaluated its selected channels, offline classification accuracy, and computation time required for each step.

The purpose of this evaluation is to determine the cost-benefit relationship of using each method for channel selection in pediatric EEG datasets. Benefits were evaluated by comparing the classification accuracy using the selected channel subset to the classification accuracy using all channels. Costs were evaluated by assessing the computation time required for each algorithm. This cost-benefit analysis which quantitatively compares different wrapper methods for channel selection is a first in the BCI literature. This work also provides a first-of-its-kind demonstration of how channel selection can improve the performance of pediatric users of MI BCI systems.

II. METHODS

A. Datasets

Channel selection methods were performed offline using calibration data from two currently unpublished pediatric BCI studies involving typically developing children aged 5-17 years. The first study [29] provides data from a single session, DS1. EEG data for DS1 were collected using the DSI24-C headset with 19 dry active EEG electrodes (Wearable Sensing, San Diego, USA). The second study [30] provides data from the same participants for two sessions with different experimental conditions, DS2 and DS3. DS2 and DS3 were collected in a random order from different participants, with some completing DS2 first and others completing DS3 first. EEG data for DS2 and DS3 were collected using the g.tec gUSB amplifier with 16 gSCARABEO wet active electrodes (g.tec medical engineering GmbH, Austria). Further details about each dataset are included in Table I. Both studies received ethics approval from University of Calgary's Conjoint Health Research Ethics Board (REB22-0044) and (REB21-1883), respectively.

DS1 and DS2 were collected using a standard calibration protocol. In this protocol, the participant was presented with two squares. They were asked to imagine squeezing their left hand when the left square became enlarged, and imagine squeezing their right hand when the right square became enlarged. Each imagined action was continued for 9 s (6×1.5 s) or 12 s (6×2 s) for DS1 or DS2, respectively. Participants were asked to relax when both squares returned to their default size for a break of 6 s between consecutive imagined actions. After the first round of left and right, feedback was provided to the participants based on an initial classification on the first 12 epochs of recorded data. When a left or right hand action was predicted, the participant would be provided with visual feedback showing

the corresponding square becoming saturated with colour, returning to its neutral grey colour when the participant was asked to relax. This was repeated a total of 9 times for each imagined action in DS1, and 10 times for each imagined action in DS2 and DS3.

DS3 used an experimental, gamified, calibration protocol. In the gamified protocol, the participant is presented with an animated monkey in the middle of the screen. They were asked to imagine squeezing their left/right hand to "catch" bananas as they fall from the top of the screen on the left or right side. Feedback was provided to the participants by having the monkey move left or right to catch the bananas. Only the visual aspects of the training paradigm are different between DS2 and DS3.

B. Signal Processing and Classification

EEG data from each dataset was saved in labelled epochs of 2 s for DS1 and 1.5 s for DS2 and DS3. Epochs were filtered with a 5th order Butterworth bandpass filter which passed frequencies of 5-30 Hz. This step was only completed once for each session to minimize the amount of redundant processing within the wrapper.

Each filtered epoch ($n \times m$) was converted into its covariance matrix ($n \times n$), where the n is the number of available channels and m is the number of samples in each epoch. The tangent space vector representation of the signal ($n(n+1)/2$) was calculated based on the covariance matrix. This process is explained in detail by Barachant et al. [31]. The tangent space representation was then classified using a logistic regression classifier. 5-fold cross validation was used to assess the offline classification accuracy. This process, from the filtered epochs to the classification result, is what is being wrapped by each of the four channel selection algorithms.

C. Channel Selection Algorithms

This work tested four different channel selection methods, SFS, SBS, SFFS, and SBFS. These algorithms were chosen, because they all have precedent of being used successfully in the BCI literature. Forward algorithms, SFS and SFFS are detailed in Algorithm 1. Backward algorithms, SBS and SBFS are detailed in Algorithm 2. In each step, the performance of each subset was assessed based on the offline classification accuracy using 5-fold cross validation. All algorithms were tested on the same folds of cross-validation data to ensure consistency. The algorithms were implemented with a stopping criterion $\varphi = -1$ so that they would not stop if their accuracy declined. This was done to ensure that all subset lengths would be tested for each algorithm. These algorithms differ from some previous implementations in that they use memory to store the current best subset S^* . So that having a stopping criterion φ which is too low does not negatively affect performance. In the case of SFFS and SBFS, memory is also used to keep track of previously tested sets, to reduce redundant calculations. Multiprocessing was implemented to check multiple potential channel subsets at a time. Computations were performed on a modern desktop

Algorithm 1 Sequential Forward Selection (SFS) and Sequential Forward Floating Selection (SFFS)

Require: X : Input feature matrix

$f()$: Evaluation function which calculates the accuracy based on a given channel set

φ : Stopping criterion

Ensure: S : Set of k selected features

$S \leftarrow \emptyset$ $\triangleright$ Initialize set of selected channels

$\theta^* \leftarrow 0$ $\triangleright$ Initialize best set accuracy

$U \leftarrow \{1, 2, \dots, n\}$ $\triangleright$ Initialize set of candidate channels

while $k < n$ **do**

$J \leftarrow \emptyset$ $\triangleright$ Initialize set of scores for candidate channels

for $j \in U$ **do**

$S' \leftarrow S \cup \{j\}$ $\triangleright$ Add channel j to test set, S'

$J \leftarrow J \cup \{f(S')\}$ $\triangleright$ Add evaluated accuracy of S'

end for

$j^* \leftarrow \arg \max_{j \in U} J$ $\triangleright$ Get highest scoring channel

$\theta_k \leftarrow \max_{j \in U} J$ $\triangleright$ Get accuracy from including j^*

$S \leftarrow S \cup \{j^*\}$ $\triangleright$ Add j^* to selected set

$k \leftarrow k + 1$

$U \leftarrow U \setminus \{j^*\}$ $\triangleright$ Remove j^* from candidate set

if $\theta_k > \theta^*$ **then**

$S^* \leftarrow S$ $\triangleright$ Update best channel set

$\theta^* \leftarrow \theta_k$ $\triangleright$ Update best channel set accuracy

end if

if $\theta_k < (\theta_{k-1} + \varphi)$ **then**

break

$\triangleright$ Stop iteration

end if

if SFFS **then** $\triangleright$ Do conditional exclusion for SFFS

while $k \geq 2$ **do**

$H \leftarrow \emptyset$ $\triangleright$ Init. scores of candidate channels

for $h \in S$ **do**

$S' \leftarrow S \setminus \{h\}$ $\triangleright$ Remove h from test set

$H \leftarrow H \cup \{f(S')\}$ $\triangleright$ Add accuracy of S'

end for

$h^* \leftarrow \arg \max_{h \in S} H$

$\theta'_{k-1} \leftarrow \max_{h \in S} H$

if $\theta'_{k-1} > \theta_{k-1}$ **then**

$S \leftarrow S \setminus \{h^*\}$ $\triangleright$ Remove h^* from set

$k \leftarrow k - 1$

$\theta_k \leftarrow \theta'_k$

$U \leftarrow U \cup \{h^*\}$ $\triangleright$ Add h^* to candidate set

if $\theta_k > \theta^*$ **then**

$S^* \leftarrow S$ $\triangleright$ Update best channel set

$\theta^* \leftarrow \theta_k$ $\triangleright$ Update best set accuracy

end if

else

break $\triangleright$ Condition for exclusion not met

end if

end while

end if

end while

return S^*

Algorithm 2 Sequential Backward Selection (SBS) and Sequential Backward Floating Selection (SBFS)

Require: X : Input feature matrix

$f()$: Evaluation function which calculates the accuracy based on a given channel set

φ : Stopping criterion

Ensure: S : Set of k selected features

$S \leftarrow \{1, 2, \dots, n\}$ $\triangleright$ Initialize set of selected channels

$\theta^* \leftarrow 0$ $\triangleright$ Initialize best set accuracy

$U \leftarrow \emptyset$ $\triangleright$ Initialize set of removed channels

while $k \geq 2$ **do**

$J \leftarrow \emptyset$ $\triangleright$ Initialize set of scores for candidate channels

for $j \in S$ **do**

$S' \leftarrow S \setminus \{j\}$ $\triangleright$ Remove channel j from test set

$J \leftarrow J \cup \{f(S')\}$ $\triangleright$ Add evaluated accuracy of S'

end for

$j^* \leftarrow \arg \max_{j \in S} J$ $\triangleright$ Get highest scoring channel, j^*

$\theta_k \leftarrow \max_{j \in S} J$ $\triangleright$ Get accuracy from including j^*

$S \leftarrow S \setminus \{j^*\}$ $\triangleright$ Remove j^* from selected set

$k \leftarrow k - 1$

$U \leftarrow U \cup \{j^*\}$ $\triangleright$ Add j^* to removed set

if $\theta_k > \theta^*$ **then**

$S^* \leftarrow S$ $\triangleright$ Update best channel set

$\theta^* \leftarrow \theta_k$ $\triangleright$ Update best channel set accuracy

end if

if $\theta_k < (\theta_{k-1} + \varphi)$ **then**

 break $\triangleright$ Stop iteration

end if

if SBFS **then** $\triangleright$ Do conditional inclusion for SBFS

while $k < n$ **do**

$H \leftarrow \emptyset$ $\triangleright$ Initialize scores of candidate

channels

for $h \in U$ **do**

$S' \leftarrow S \cup \{h\}$ $\triangleright$ Add h to test set, S'

$H \leftarrow H \cup \{f(S')\}$ $\triangleright$ Add accuracy of S'

end for

$h^* \leftarrow \arg \max_{h \in U} H$

$\theta'_{k+1} \leftarrow \max_{h \in U} H$

if $\theta'_{k+1} > \theta_{k+1}$ **then**

$S \leftarrow S \cup \{h^*\}$ $\triangleright$ Add h^* to selected set

$k \leftarrow k + 1$

$\theta_k \leftarrow \theta'_k$

$U \leftarrow U \setminus \{h^*\}$ $\triangleright$ Remove h^* from set

if $\theta_k > \theta^*$ **then**

$S^* \leftarrow S$ $\triangleright$ Update best channel set

$\theta^* \leftarrow \theta_k$ $\triangleright$ Update best set accuracy

end if

else

 break $\triangleright$ Condition for inclusion not met

end if

end while

end if

end while

return S^*

processor, Ryzen 3900x (Advanced Micro Devices, Santa Clara, USA), with 12-cores and 24 threads. At each step, the time, classification accuracy, and channel subset in that step was recorded for each algorithm and each participant. To compare the effects of different stopping criteria, the accuracy improvements from each step was calculated and if the improvement was less than the stopping criterion then the list was truncated to only include entries up until that point to simulate the algorithm stopping. This was used to determine the time-performance tradeoff for each algorithm with stopping criteria from -0.15 to 0.15 in increments of 0.01. Having a negative stopping criterion enables the algorithms to continue even if a single step results in a small reduction in accuracy.

III. RESULTS

The mean accuracy of each algorithm when run to completion is shown in Table II. From the table, it is clear that all channel selection algorithms result in a 10-15 % absolute gain in mean accuracy for each algorithm compared to when no channel selection algorithms are used. SBFS is the algorithm with the greatest mean accuracy of 75 %, followed by SFFS, SBS, and SFS. The mean completion times for each algorithm on each dataset are shown in Table III. Completion times are very similar for SFS and SBS, which took the least time to complete, followed by SBFS and SFFS. In all cases, DS2 and DS3 took less time to complete than DS1. However, both classification accuracy and completion time exhibit high variability, which is captured in Fig. 1. Fig. 1 shows the relationship between classification accuracy improvement and computation time for all algorithms across all datasets. It shows that SFS and SBS have high variance in performance gain for each individual and little variance in computation time. Fig. 1 also shows that the distribution of completion time for SFFS and SBFS has positive skew with maximum completion times of 77.0 s and 53.7 s, respectively for DS1.

TABLE II
MEAN CLASSIFICATION ACCURACY (%) $\pm$ SEM

Dataset Algorithm	DS1	DS2	DS3
None	60.4 $\pm$ 2.0	58.5 $\pm$ 2.4	62.3 $\pm$ 2.7
SFS	72.5 $\pm$ 1.3	69.7 $\pm$ 1.7	72.8 $\pm$ 2.1
SBS	73.9 $\pm$ 1.3	71.3 $\pm$ 1.6	73.3 $\pm$ 2.0
SFFS	74.9 $\pm$ 1.2	72.5 $\pm$ 1.6	74.8 $\pm$ 2.0
SBFS	75.5 $\pm$ 1.2	72.8 $\pm$ 1.6	75.4 $\pm$ 1.9

TABLE III
MEAN COMPLETION TIME (s) $\pm$ SEM

Dataset Algorithm	DS1	DS2	DS3
None	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
SFS	9.5 $\pm$ 0.4	6.4 $\pm$ 0.1	6.3 $\pm$ 0.1
SBS	9.9 $\pm$ 0.4	6.4 $\pm$ 0.0	6.3 $\pm$ 0.1
SFFS	35.8 $\pm$ 1.9	23.0 $\pm$ 1.1	21.6 $\pm$ 1.0
SBFS	29.3 $\pm$ 1.3	18.5 $\pm$ 0.6	18.7 $\pm$ 0.7

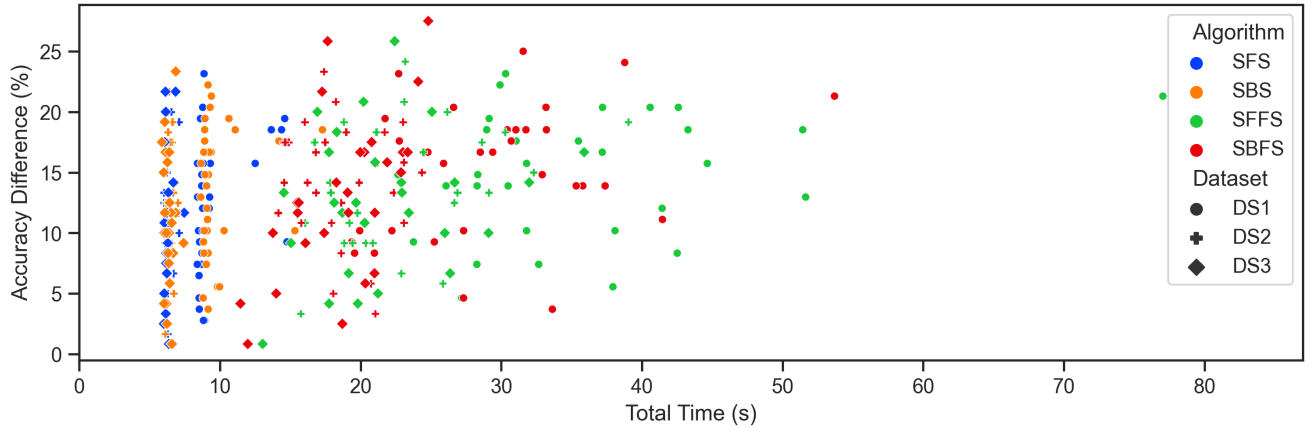


Fig. 1. Scatter plot of absolute classification accuracy difference compared to all channels v. total time for subset selection for each algorithm and each dataset.

The distributions of classification accuracy improvement from using all channels ($\theta^* - \theta_{k=m}$) are shown in Fig. 2 for all possible channel lengths. Fig. 2 also shows that the greatest accuracy gains are found with channel lengths of 5-12 for DS1 and 5-10 for DS2 and DS3, with gains waning at either end of the spectrum. With a single selected channel, the median accuracy is similar to the accuracy with all channels, but with a much greater variance. This variance wanes at larger numbers of channels.

To determine which channels were being selected for each subset size, topographic maps of the selected locations of channels for subset lengths $k = 1, 2, 4, 8$ are shown for DS1 in Fig. 3. Only SFFS and SBFS are shown, because the distributions are similar to SFS and SBS, respectively. The topographic plots in Fig. 3 show that when one or two channels are selected, electrodes C3, C4, O1, and O2 predominate. SFFS has a greater concentration around C3, C4, while SBFS has a greater concentration around O1. With a growing number of channels, the distribution of electrode selection becomes more uniform and more similar between SFFS and SBFS.

Fig. 4 shows the effect of changing the stopping criterion (φ) on the accuracy improvement and completion time of each algorithm for DS1. Accuracy and completion time are both on s-shaped curves with respect to the stopping criterion. The far left shows the case where the stopping criterion never triggers the algorithm to stop. In this case, the algorithm is able to check channel combinations of all lengths. The far right shows the case where the stopping criterion immediately triggers the algorithm to stop after the first step. Moving right, time to completion begins to decline at $\varphi = -0.15, -0.07$, and -0.05 for SBFS, SFFS, and SFS/SBS respectively. At these points, each algorithm has already achieved their maximal accuracy. The decline in accuracy continues to lag behind the decline in computation time until they both reach the lower bound of the curve. At $\varphi = 0$, all algorithms offer comparable improvements around 8 % while only requiring 4 s for computation. At the far right, SBS and SBFS offer greater performance than SFS and SFFS.

IV. DISCUSSION

Channel selection algorithms SFS, SBS, SFFS, and SBFS all offer a means of improving classification accuracy for pediatric users compared to using all channels. This supports what has been seen in the literature with adult users. Some algorithms perform better than others, but the differences are marginal (2.5-3% difference between SFS and SBFS) compared to using all channels for classification (10-12% difference between SFS and no selection). Improvements in classification do not affect all individuals equally, as shown by the variance in Fig. 1 and Fig. 2. However, because of the use of memory in these selection algorithms, a subset will never be selected which has worse accuracy than when using all channels. Each algorithm frequently selected C3 and C4 when subset length was limited. Selecting these electrodes, located over the motor regions, indicates neurophysiological plausibility of the selected channels. The frequent selection of O1 and O2 is less expected, but could be because these electrodes far from the motor cortex contain less mutual information with C3 and C4 and are therefore more beneficial. The relative uniformity of the selected channels at larger subsets suggests that after the most salient electrodes are selected, the role of the channel selection algorithm is to leave out channels which do not benefit the classifier. In this evaluation of typically developing children, this is likely leaving out channels with lower signal quality. For individuals with differences in head shape or brain structure, this could be beneficial, because it would leave out the channels with poor contact quality without requiring additional time to adjust or adapt the headset.

The floating algorithms SFFS and SBFS performed better than their non-floating counterparts SFS and SBS. This is because these algorithms have the ability to avoid the nesting effect by removing channels which were previously added (SFFS) or adding channels which were previously removed (SBFS). This is important, because proximal channels contain mutual information, and can become redundant if nearby channels are added. However, these algorithms add a large

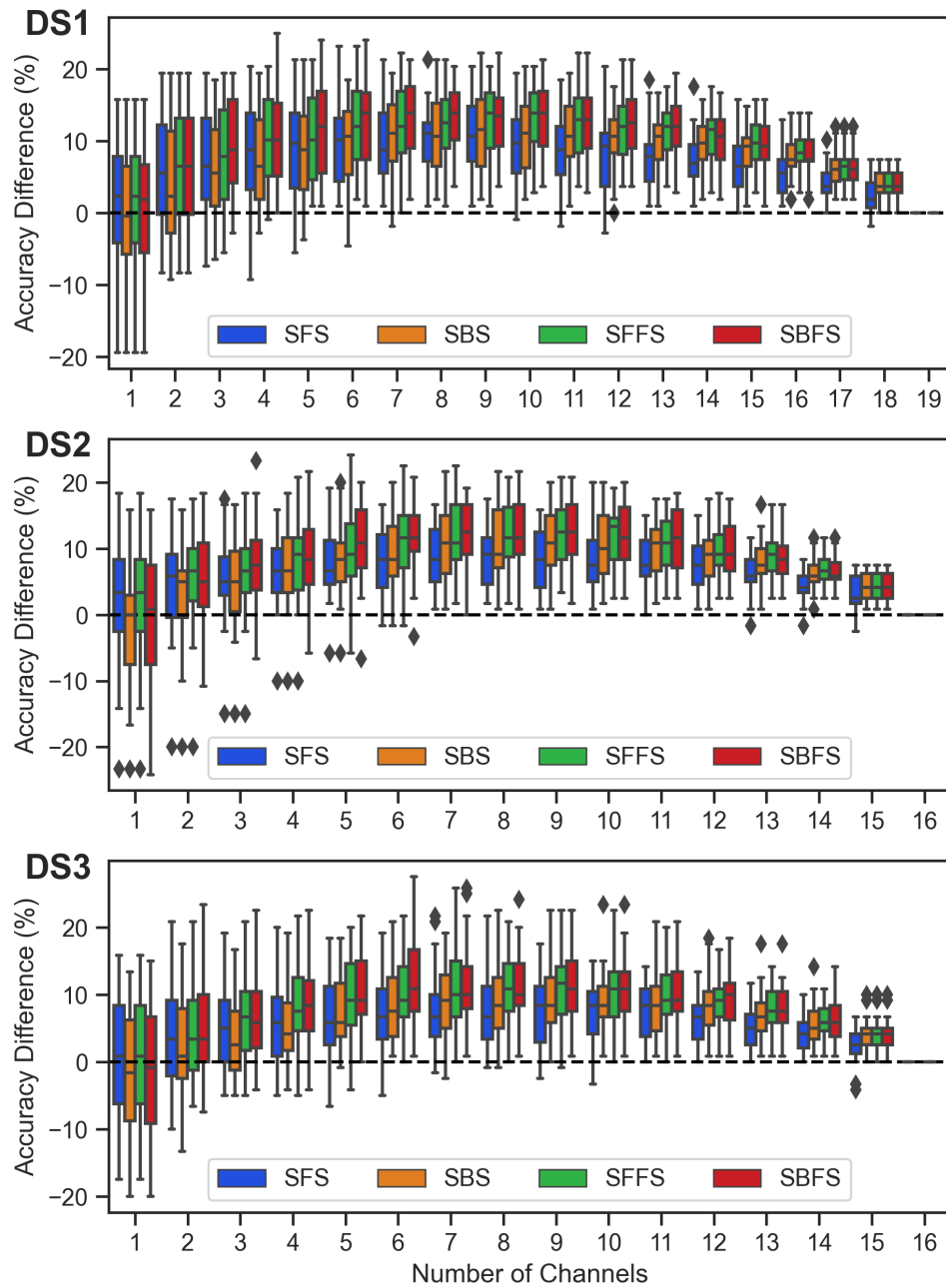


Fig. 2. Classification accuracy of subsets computed with channel selection compared to classification accuracy using all channels for three datasets DS1, DS2, and DS3.

and variable time cost, so should only be used when users will tolerate a break between calibration and the BCI task. Backward selection algorithms SBS and SBFS performed better than their counterparts SFS and SFFS. The reason for this is presently unclear, but it provides a small amount of evidence to choose backward methods over their forward counterparts. In particular, SBFS should be considered over SFFS, because it results in both higher accuracy and lower completion time. Changing the stopping criterion offers another option for limiting the completion time of the algorithm without a proportional decrease in performance.

To our knowledge, this is the first work to test channel selection methods for pediatric users of BCIs. It is a clear demonstration of the impact that state-of-the-art methods can have on pediatric BCI. This is also the first work to demonstrate and compare the time-accuracy relationship of different channel selection methods for either pediatric or adult users. Finally, this work demonstrates how adjusting the stopping criterion for these algorithms can influence the cost-benefit relationship of each tested algorithm. Based on these assessments, we can make some recommendations on how these algorithms can be applied to future BCI systems.

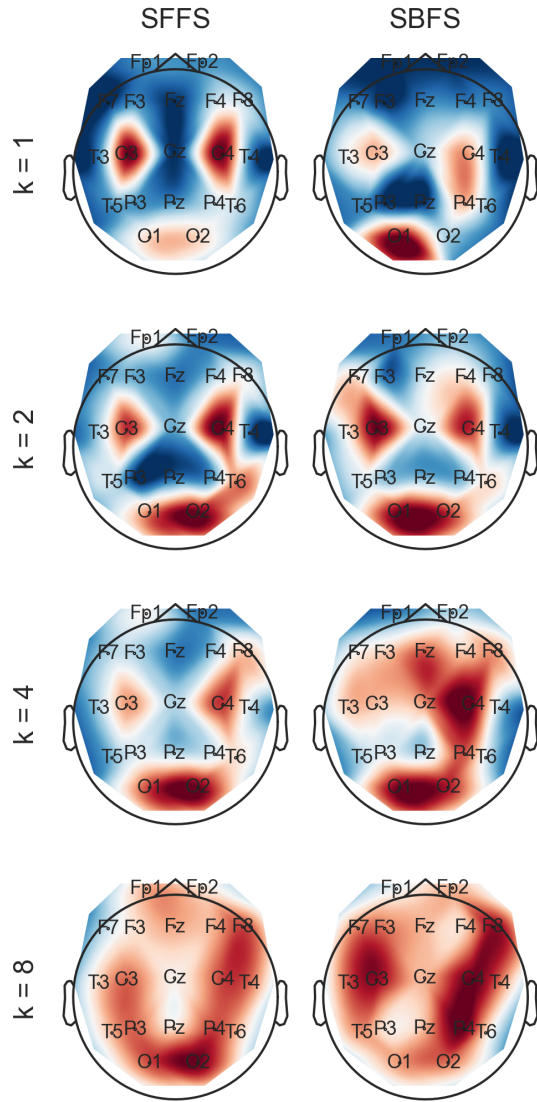


Fig. 3. Topographic plots showing the distribution of selected channels for subset sizes $k = 1, 2, 4, 8$. Red indicates channels which were commonly included in the subsets of each length.

The key takeaway is that using any of these channel selection algorithms is better than nothing. The demonstrated accuracy gains can meaningfully improve the usability of a BCI system, particularly for children. If time is constrained or computational resources are limited, consider using SBS. To further reduce computation time, use a stopping criterion of 0, which in our testing offered a mean accuracy improvement of 8 % in less than 4 s. If you have one minute or more, then readers may consider using SBFS to get the highest offline classification accuracy. These recommendations hold in situations where multiprocessing is possible and the number of available threads is greater than the number of channels. This optimal case enables each possible combination to test can be allocated to its own thread. In cases where there are more channels than threads, computation will be slower, but still worthwhile.

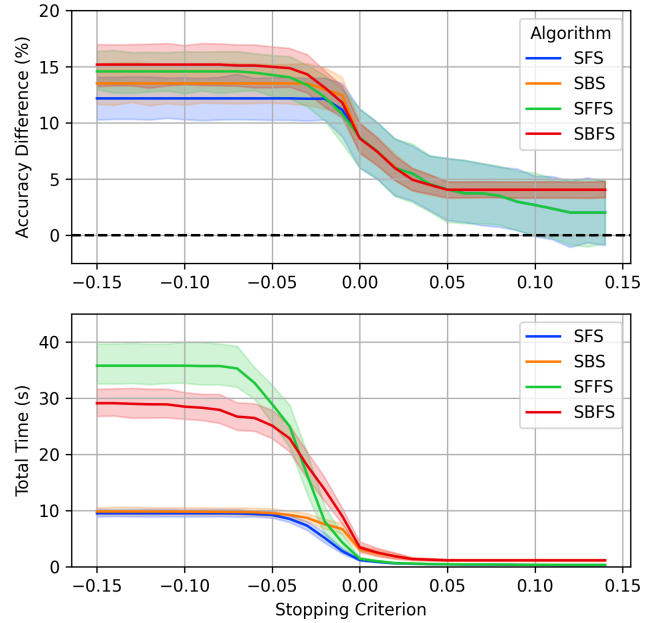


Fig. 4. Effect of changing stopping criterion on time and accuracy for DS1. Bold lines and shaded areas indicate the mean and 95% confidence interval, respectively.

A limiting feature of this work is that it measures accuracy offline. In this way, it does not necessarily capture the online performance of these algorithms, and performance could be disproportionately impacted if signal quality degrades on one of the selected channels. Future work should investigate the efficacy of these algorithms in online systems, as well as with other BCI paradigms. Future work should also evaluate these algorithms for use with children with QCP and other complex needs, as this population has the most to gain from improved BCI performance through personalization.

V. CONCLUSION

In this work, we investigated four wrapper-based channel selection algorithms on three pediatric motor imagery datasets. All channel selection methods found subsets which performed better than using all available channels. This indicates that channel selection is beneficial for typically developing children using MI BCI. The algorithms all prioritize areas of high salience which match expectations for users with typical motor organization. SBFS provides the greatest boost to performance of all the tested methods. We recommend BCI practitioners consider adding this method to their pipelines for improved personalization and performance, with little computational cost. Where time is limited, we recommend using SBS, possibly with an increased stopping criterion, to further reduce the computational cost of the algorithm. These results are evidence of how much pediatric users of BCIs have to gain from state-of-the-art techniques such as channel selection. This work motivates the further application of state-of-the-art techniques to further improve access to personalized and performant BCIs for pediatric users.

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